

Intelligent scheduling and planning systems for telecommunications resource management

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This paper discusses the importance of resource management in the telecommunications industry. It argues for intelligent resource management systems (IRMSs) that are capable of providing flexible and dynamic resource management. It introduces the main requirements and benefits of an IRMS and describes the technical basis of current research activities which are leading towards the development of constraint technology based IRMS platforms.

1. Introduction

The dream of integrated broadband communications, or more generally that of information, communications and entertainment (ICE) systems, is rapidly turning into a reality. The growth of a multiservice, multivendor environment will provide opportunities for service providers and network operators to offer customers sophisticated value-added services.

In order to exploit fully the benefits of the emerging ICE systems, the various actors will need to plan and use their available resources (henceforth referred to as 'resource management') effectively to meet growing user expectations in an ever more competitive and rapidly changing international market. Effective resource management will have a direct influence on business performance and profitability, and have a major impact on the advantages provided to customers. Accordingly, resource management has been identified as a key differentiator by many telecommunications companies and relevant R&D programmes [1–5].

Technical approaches undertaken by many existing resource management systems are restrictive and incapable of providing satisfactory and comprehensive solutions to problems at hand. It must be recognised that existing solutions to resource management will not meet the business demands of the future ICE market. There is a clear need for intelligent resource management systems capable of providing flexible planning, scheduling and allocation of resources of modern information networks. In addition, it is becoming recognised that resource management technology will be an important driver for new business opportunities. This paper is structured as follows. In section 2, telecommunications resource management concepts and requirements are discussed and examples of typical resource management tasks are given. In section 3, the main

requirements of an intelligent resource management system and its conceptualisation in terms of constraint technology are introduced, and this is followed by a brief summary and references.

2. Resource management

2.1 Definition

A concrete definition of what constitutes resource management is desirable, but almost certainly unfeasible. Indeed, what constitutes a resource is also probably best left undefined (in any rigid sense), for any prescriptive list will almost certainly be inadequate and will quickly become outdated. However, as an indication, the following should be considered to be examples of resources — 'people, transmission bandwidth, switching equipment, monitoring equipment, strategic knowledge, and customers' [6].

Resource management is concerned with, among other things:

- maintaining an inventory of resources,
- managing supply and installation of new resources,
- the allocation and de-allocation of resources,
- planning and scheduling of activities and resources.

This covers a broad spectrum of the tasks associated with managing an enterprise and therefore includes tasks which, while important, will be considered peripheral to the main thrust of a telecommunications business. Tasks of principal importance will involve a finite set of resources, a finite set of resource requests and a number of application-specific constraints. These tasks will require the resources to be matched with resource requests in such a way as to minimise some predetermined cost function. Thus the maintenance of an inventory of resources, for example, is a 'means to an end' and not an issue of central importance.

2.2 Importance of resource management

Many companies have now recognised a critical need to develop advanced resource scheduling and planning systems in order to meet their customer commitments improve their quality of service, and reduce their operational costs. In the case of large telecommunications companies which provide services to the international community, planning and scheduling systems form an essential part of the process of responding quickly and efficiently to the requirements of customers. These involve the installation or maintenance of equipment and networks, or the automatic allocation of transmission capacity (e.g. video, data, telephony), which have resulted from a customer's request for service. In particular resource management must be able to:

- efficiently implement current working practice,
- make best use of the existing resources,
- act as a driver for a new service offerings.

Hitherto, most large companies have employed resource management techniques to address only one of the above points, i.e. to improve the efficiency of current practice. It is now clear that customers are demanding greater flexibility - a flexibility that existing resource management techniques are unable to deliver. Thus an impasse has been reached - existing solutions to resource management are at odds with the demands of the new business culture.

But it is crucial that the technology goes beyond 'just catching up'. Resource management techniques must be able to satisfy fully the needs of the business, and in addition they should be one of the driving forces behind new service offerings. Extensive experimentation with advanced information processing (AIP) techniques [7] by BT and their university collaborators have identified such techniques as a rapid way of prototyping new management techniques and evaluating their implications. Thus AIP techniques/systems can be used as an environment in which new approaches to resource management can be

'manufactured', safe in the knowledge that they can be deployed rapidly in the business.

2.3 Resource management problems in the telecommunications industry

A number of typical resource management application areas within BT are discussed in the following sections. The problem areas, the existing solutions and the drivers for future solutions are identified.

2.3.1 Network management

Network operators manage a variety of networks (video, telephony, data, etc) in many locations, but the efficient allocation of network resources is a problem common to all of these networks. Existing solutions to these problems vary according to network size and usage patterns. Future solutions must satisfy the following requirements:

- more flexibility,
- better network utilisation,
- lower cost,
- faster allocation.

It is clear from initial studies on current networks that more sophisticated approaches to resource management are required. This need will become more urgent with the expected increase in the complexities of future information networks. Applications of current AIP- based scheduling and planning techniques to this problem have been successful and already meet many of the business needs. However, several objectives still cannot be met with current technology:

- reacting to network failure and changes in configuration by real time dynamic rescheduling,
- testing the feasibility and the impact of an allocation — this is crucial in situations where an immediate response is required indicating whether an allocation is feasible or not.

In the following, the role of resource management in some of the important aspects of network management will be described.

Network routing

Network routing is the process of deciding the path through a network a given transmission path should take. This path may be from a single source to a single destination, or it may be a single source to multiple destinations. Requests to perform routing may arrive some time in

advance (in which case it is more like a booking function) or they may arrive and must be serviced, in real time. The primary resource to be managed is the transmission network, and more importantly the available capacity on that network. Requests to provide a transmission path can be viewed as a request to be allocated some of the network capacity resource.

Network routing occurs in all telecommunications networks in a variety of ways. To provide greater insight, some of the application areas of network routing are considered here.

- Case study — visual and broadcast services

BT supplies and manages sound and vision networks used, primarily, by major UK broadcasters (television and radio) and the broadcast support industry. In addition, BT provides international videoconferencing facilities and a satellite data service. The network is supported by a mixture of digital and analogue equipment over a variety of transmission media (microwave, copper, optical fibre, etc).

For example, a television programme being broadcast from Granada studios is fed directly to the nearest network switch (at Manchester) and from there it is routed across the country until it reaches the regional distributors who are then responsible for transmitting the programme to people's homes. In addition to routing preplanned television broadcasts, BT also supports ad hoc requests for capacity. Such requests are often placed when there is a need, for example, to receive news coverage from an outside broadcast unit. Many of the requirements for an automatic routing system will be common to other BT network systems such as managed bandwidth services and videoconference services running over the global video network.

- Current approach

Network booking is currently performed semiautomatically by BT's booking staff. There are two types of booking performed at the moment. Large users of the video and audio networks, such as the television companies, have much of the network reserved for their use only, and currently stipulate exactly how these network resources should be arranged to support the services that they require. Smaller users, with ad hoc requests, are allocated network capacity as and when required (where possible). BT is currently in the process of a major programme to upgrade its video and audio networks and to rollout automatic switch controllers in place of

the existing system. Such controllers raise new possibilities concerning the use of automatic booking systems.

- Requirements for future solutions

When all the national switching centres have automatic switch controllers, BT will be in a position to completely reappraise the way in which the network is managed. Following a prototyping experiment, a list of requirements has been prepared (see below). It should be noted that this list is tentative and this is very much an area where technology can provide business drivers.

- BT operates both a video network and an audio network. It is a requirement that, wherever possible, solutions for one type of network be equally applicable to the other.

- The way in which a transmission is routed will not be made visible to the customer.

- Customers should be able to contact BT booking staff, via telephone or remote terminal, and be able to make requests about bookings and receive responses within a very short time (seconds). Such queries could be:

‘Can I have a route between A and B, at time T?’

‘At what times can I have a route between C and D?’

- Requests to schedule broadcasts are generally received well in advance of transmission. It should be possible to take a number of such broadcasts and automatically determine the best way to route them.

- It is often the case that there are changes to broadcast schedules right up to the point of transmission. It should be possible to produce alternative booking schedules in the light of these changes.

- If there is a network failure it should be possible to re-route affected services automatically.

Network design

Network design is the (conceptual) assembly of network elements to meet some network specification. The network elements used can be viewed as resources and the assembly of these resources to produce a desired network as a resource management task. Initial requirements for a network (such as traffic volumes, geographical location, cost constraints) are converted into a plan of a network that meets customer needs. Many key decisions are made in this process, a number of which may be automated or aided by resource management decision support tools. Such decisions include:

- cable layout,
- switch placement,
- switch and cable dimensioning,
- transmission bandwidth.

- Current approach

The BT network design centre programme is investigating a wide variety of network design techniques and tools. These include a decision support tool that aids the user in the cost analysis of network designs, and a system which converts the schematic diagram of a network into a physical realisation with appropriately dimensioned hardware [8]. Elsewhere conventional extrapolation algorithms (some in the form of commercial off-the-shelf products) are used to perform capacity planning. Switch siting is performed manually.

- Requirements for future solutions

A large body of work is already under way within the network design programme. Development of decision support tools for networks design and planning will be greatly assisted by AIP constraint based design and optimisation techniques, and in fact AIP stochastic techniques are being employed in parts of the programme [9].

2.3.2 Workforce management

Traditionally telecommunications companies have large maintenance workforces. The close supervision of field engineers and technicians represents a large management activity. Maximising their productive time and optimising their availability are crucial to the financial viability of the company and large sums of money have been invested in suitable computer-based scheduling systems to address this problem.

The existing workforce scheduling systems attempt to match a set of engineers to a set of jobs and have shown significant improvements in achieving the objective of 'getting the engineers with right skills, to the right place, at the right time, with the right stores'. The current systems are mostly designed for predictive (look-ahead) scheduling. They operate largely in a mode which allows little scope to react efficiently to problems on a real time basis. Many of these systems are based upon traditional operations research techniques [10] which make them difficult to maintain and inefficient for large scale problems. Some of the other main limitations of existing systems include:

- they do not support large scale real time scheduling or dynamic rescheduling,
- they do not make use of new promising AIP constraint scheduling/planning techniques,
- they do not provide facilities for running 'what-if' scenarios for long-term workforce and resource planning.

WM case study — in 1988, BT's work management (WM) programme was set up to look for major improvements in the work management activity. WM can involve dispatching engineers for:

- repair' (either of customer premises equipment (CPE) or network elements),
- 'installation' (again installation of either CPE or network elements, or a combination of both).

It should be noted that in the case of CPE it is clearly important to indicate in advance the time the engineer will arrive. There are a variety of service levels, and customers with the higher levels must be dealt with within shorter time scales. Also to provision an analogue private circuit across the UK requires installation of CPE at either end as well as installation or modification at the exchanges through which the circuit is established.

Current approach — the main deliverable from the work management programme has been the Workforce Management System (WMS) which began field trials in 1991. The main aims of the WMS are:

- to provide electronic communication between engineer and a central system,
- to facilitate significant improvements in the decision process with regard to resource allocation and control.

The papers by Laithwaite [2] and Simpson et al [10] give a detailed technical account of the WMS.

Requirements for future solutions — WMS developers are actively planning enhancements to the current WMS in dealing with more constrained and dynamic workforce management problems. These include:

- Better reactive scheduling

Many of BT's business customers have contracts requiring response within 5 hours. Although the current real-time algorithm in the WMS is capable of supporting such a method of working, it must manage changes in a more controlled manner to preserve previous solution stability and optimisation goals.

- More intelligent scheduling of engineers to complex jobs

Some of BT's network products can only be serviced by a group of engineers working in some predefined sequence. Examples of this type of work include circuit line-up across networks and installing major customer premises equipment. The allocation of engineers to these jobs and specific reactive scheduling procedures, in cases where these jobs are affected by domain changes, are not currently supported by WMS. Future work management systems should be able to support these scenarios.

- Support for more heuristic control

Users of WMS would like to see more heuristic control in the form of 'rules' that can be changed relatively easily.

- More efficient allocation and resource planning

With manpower being one of the largest cost components within BT, there will always be pressure for more efficient deployment of field people. Furthermore, availability of the right level of manpower and for that matter other resources, e.g. stores, to meet business needs is a fundamental feature required of future WMSs. There is a growing need to predict and plan for future activities and required resources based on business objectives and historical information.

- More specific appointment scheduling

Currently most customers are given appointment times that specify morning or afternoon. There is a recognition that more precise appointments (e.g. at some time between 1000 and 1100 hrs) would be desirable but current scheduling techniques are not sufficiently powerful.

In summary, telecommunications industry resource management requirements focus on network and service allocation, workforce scheduling, and resource

monitoring and planning. All require advanced reactive scheduling and planning systems capable of responding quickly to changing environments. They must have the capacity to co-ordinate externally imposed time constraints which may arise both from preferences (priorities) or other predetermined criteria and from changing circumstances.

3. Intelligent resource management system

3.1 Resource management — an AIP perspective

The main decision-making objectives in resource management applications are:

- efficient utilisation of resources,
- rapid response to demands,
- close conformity to deadlines.

The basic constraints of a resource management problem relate to the resources available and the restrictions affecting the order in which the required tasks can be performed. Resources typically require constraints of two sorts, quantitative and qualitative. Tasks are described in terms of resource requirements, including duration, release and due dates. A solution to a resource management problem allocates resources to tasks and orders the requisite tasks, subject to the constraints posted. If the set of tasks to be scheduled does not change over scheduling and/or execution time, the resource management problem is called static or predictive. If new tasks arise over scheduling and/or execution time, the resource management problem is called dynamic or reactive.

Recently, AIP techniques have been used to address some of the critical problems which arise in resource management applications, such as efficient search execution, integration of scheduling and planning, and dynamic scheduling (or reactive scheduling). Among the AIP methods, approximation techniques such as genetic algorithms, and simulated annealing, and constraint solving techniques such as domain filtering, tree search [7] and no-good learning [11] are by far the most popular approaches to complex resource management problems.

A significant proportion of current AIP-based resource management R&D work has viewed scheduling, satisfaction problems (CSPs) [12]. Despite much reported success in the formalisation of the problem, this view suffers from well-known limitations of classical CSP frameworks [13, 14]

These limitations are to do with constraint representation (defining a set of complete and rigid constraints) and constraint manipulation (generating only one or a small number of solutions). The repercussion of

these limitations are that some important features which are central to IRMSs are missing in the classical CSP view of resource management. These features, which are described in the next section, are:

- handling user or environment imposed changes to the problem at hand,
- richer constraints representation,
- capturing and reasoning with the domain incompleteness or imprecision [14].

3.2 IRMS requirements

An intelligent resource management system is a resource scheduling and planning system which must meet a number of important requirements.

- *It should have the ability to respond to a changing environment without recourse to time-consuming resource scheduling and planning activities.*

As described earlier, many real-life resource management problems are dynamic in nature and as such cannot be tackled by classical CSP approach. A changing resource management problem can be viewed as a sequence of CSPs which need solving in a flexible and incremental fashion.

When the resource management system is operating in an unstable or changing environment, the system must incorporate techniques such as solution maintenance and incremental schedule modification [15]. Two important principles have to be taken into account in developing intelligent constraint-based reactive/dynamic resource management systems:

- the principle of ‘incrementality’, which states that constraints, resources and activities can be added and removed from the schedule at any point in time,
- the principle of ‘locality’, which states that each constraint relates to as few resources and activities as possible.
- *It should have rich modelling capabilities in order to represent the inherent structure of the problem domains and capture user-defined preferences and reason effectively with them.*

Many real-life resource-management problems have an inherent hierarchy which ought to be represented and exploited in solving such problems. The classical CSP formulation of resource-management problems is flat and requires prior declaration of a complete set of variables, domains and constraints. Furthermore, the classical CSP does not incorporate natural

representation and handling of user-defined preferences (priorities). They are either avoided or hard-wired as set of cost constraints in the problem representation. The first renders the problem under-constrained whereas the latter causes an overconstraining of the original problem [14].

However, a hierarchical representation of resource management problems would further complicate the representation task as the set of variables and their associated domains and constraints are not fully declarable a priori and become available or more precisely defined during the scheduling and planning activities. In addition, the introduction of preferences in the problem representation provides a whole range of possible solutions to the problem, i.e. from highly preferred solutions to least preferred ones. Naturally, the introduction of preference would take place on constraints and local or global assignments of variables.

- *It should be able to tackle distributed resource-management problems.*

IRMS will have to work in a distributed resource scheduling environment where often not all actors’ resource requirements or constraints can be satisfied due to the competition for scarce resources. IRMS, therefore, needs to incorporate negotiation strategies for constraint relaxation and approximation.

- *It should allow for a mix of AIP techniques to be employed.*

Many industrial resource management problems are large-scale constraint-optimisation problems. Experiences have shown that while constraints satisfaction techniques (such as tree search and domain filtering) are effective in dealing with the underlying constraints of the applications, the approximation techniques (such as simulated annealing) are more effective in solution optimisation process. A mix of these techniques is essential to solve resource-management problems [15]. This suggests that IRMS must observe the principle of ‘integrability’ (which states that different constraint handling mechanisms and optimisation techniques are able to co-operate and share data).

- *It should be sufficiently flexible to deal with knowledge incompleteness or imprecision.*

A problem description could be incomplete or imprecise for many reasons, e.g. data imprecision or uncertainty, data incompleteness. A classical CSP formulation often assumes a complete description of the problem and cannot deal adequately with incomplete descriptions [14].

- *It should provide a friendly user interface (UI) and allow for flexible user interaction.*

The provision of a friendly UI and the capability for two-way communications are as crucial as any of the above described requirements. Conveying information to the user while search for a solution is underway and allowing the user to interact with the system in order to guide and influence the search or tackle the bottlenecks are extremely important.

- *It should provide facilities for predicting and planning for sufficient quantities of resources.*

It is crucial to have sufficient resources to achieve the objectives which control and/or influence the output of IRMS. To predict accurately sufficient levels of resource, it is necessary to have access to historical data, have sufficient knowledge of the relationship between the IRMS and other systems, and have powerful planning and simulation tools which are capable of interpreting knowledge and solving problems at different levels of abstraction.

The challenge is to integrate the above requirements into an AIP-based resource-management problem-solving platform which must also address more traditional resource scheduling and planning problems. Among the aims is the desire to introduce robustness and time-critical response into such a platform which must be designed to explore feasibility and optimisation [16].

3.3 AIP constraint technology for IRMS

The constraint-based approach to IRMS models dynamic change as dynamic constraint satisfaction where constraints can be asserted or retracted incrementally, reflecting a change of the world [17]. The propagation and dependency-directed backtracking aspects of constraint solving enable partial solutions to be revised. In particular, the preferences between constraints enables an opportunistic 'opt-out' in pursuit of new goals. Under the dynamic constraint-based view, by retaining knowledge of the search space explored and the remaining opportunities, there is no longer a distinction between prediction and reaction. The former simply becomes a continuation of search or conversely reaction is considered as a series of predictions. If a distinction is required, separate heuristic mechanisms for prediction and reaction may be used. This separation approach is taken by OPIS [18].

Because the constraint-based approach puts the emphasis of the resource management on dynamic constraint solving, the task at hand can be seen as being predominantly one of maintenance of the solutions rather than creation. This view is taken by some AIP-based systems such as 'S2' [19], where the same mechanisms are

used for predictive and reactive scheduling and the technology of ATMS [20] is employed to enhance prediction and action.

The constraint-based approach can further be embedded within a distributed (multi-agent) resource management framework where each agent exhibits the properties of opportunism, reaction and constraint maintenance. In this framework, agents run concurrently; to work towards a global solution. The distributed framework can be structured hierarchically (e.g. 'DAS' [21]) where agents can delegate tasks and send commands to junior agents along the hierarchy. Alternatively, it can be market-structured in the sense that any macroscopically generated pattern is self-organised depending on the constraint interaction [22].

The constraint-based approach to resource management provides a unifying view of previous approaches to resource planning and scheduling [24]; it is also conceptually simple to develop and implement. However, to enhance further the efficiency of the constraint-based approach, it is necessary to address three hard problems.

- Finding a (global) solution may generate a huge search space of constraints, many of which may be irrelevant to the current (local) process. There will be, in this circumstance, a heavy and partly unnecessary burden on constraint propagation. A strategy which can localise the task of constraint propagation will reduce the burden significantly, and contribute essentially to the viability of the approach.
- Resource scheduling and planning may involve many disjunctive constraints, some combinations of which may lead to an incorrect schedule. An intelligent back-tracking strategy must be developed to minimise the amount of redundant work.
- Resource scheduling and planning may involve many actions which compete concurrently for resources. Managing action interferences correctly and efficiently is essential to reduce the search space of scheduling.

Solutions to these problems and many others which arise from addressing IRMS requirements are now the subject of current research activities in the areas of reactive scheduling and resource planning. The aim is to develop novel techniques for solving BT's large-scale constraint optimisation problems. These include a variety of stochastic search techniques [24], analytical methods for problem difficulty detection [25], constraint satisfaction and optimisation techniques [11, 13, 25, 26], and extending existing or emerging constraint-technology scheduling and planning platforms [10, 15, 24, 27].

4. Conclusions

Effective resource management has a major impact on telecommunications network performance and profitability, significantly contributes to the advantages provided to its customers. Resource management for future BT ICE services requires an intelligent resource management system capable of planning for required resources and efficiently producing and intelligently maintaining schedules for flexible resource management in a dynamic world. The necessary AIP technology for developing such a sophisticated IRMS is investigation at BT Laboratories.

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